

1. Consider the following example of studying the relationship between test scores and class sizes in primary schools:

$$TestScore = \beta_0 + \beta_1 STR + u. \quad (1)$$

We consider three different scenarios:

- (1) The kids are randomly enrolled to nearby primary schools.
- (2) The kids are selectively enrolled: the kids with high ability will be enrolled to good schools and those with low ability will be enrolled to bad schools.
- (3) The kids' parents decide which schools to apply. In the case that the number of applicants exceeds the vacancy of the school, a random balotting will be implemented.

Discuss whether the OLS Assumption 1 is likely to hold for each scenario.

2. (a) Explain the measure of fit: R^2 .
(b) Illustrate the meaning of explained sum of squares (ESS) and total sum of squares (TSS) using a diagram.
3. (a) Show that the sum of the OLS residuals $\hat{u}_i$ ($\sum_{i=1}^n \hat{u}_i$) is equal to zero by using the first-order condition (FOC) of the OLS estimators.
(b) Show that the average of the OLS predicted values $\hat{Y}_i$ ($\frac{1}{n} \sum_{i=1}^n \hat{Y}_i$) is equal to the sample average of Y ($\bar{Y}$).
(c) Show that if the OLS estimator $\hat{\beta}_1 = 0$, then $R^2 = 0$.
4. A random sample of 100 20-year-old men is selected from a population and these men's height and weight are recorded. An OLS regression of weight on height yields:

$$\widehat{Weight} = -79.24 + 4.16 \times Height, \quad R^2 = 0.72, \quad (2)$$

where *Weight* is measured in pounds and *Height* is measured in inches.

- (a) What is the regression's weight prediction for someone who is 64 inches tall? 68 inches tall? 72 inches tall?
- (b) A man has a late growth spurt and grows 2 inches over the course of a year. What is the regression's prediction for the increase in this man's weight?
- (c) Suppose that instead of measuring weight and height in pounds and inches, these variables are measured in centimeters and kilograms. What are the OLS estimates from this new centimeter-kilogram regression? How about the R^2 ?